

Network Troubleshooting and Diagnostics Lab

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1 Project Overview

This project demonstrates practical network troubleshooting and diagnostics techniques using Cisco Packet Tracer. The lab simulates common network issues such as incorrect IP addressing and gateway misconfigurations, followed by systematic diagnosis and resolution using networking tools and connectivity testing.

2 Objectives

- Simulate common network connectivity issues
- Diagnose IP configuration and gateway problems
- Perform network testing using troubleshooting tools
- Restore network communication by correcting configuration errors
- Develop practical troubleshooting skills for enterprise environments

3 Network Topology

The network consists of:

- 1 Switch
- 3 PCs
- Static IP addressing configuration

All devices were connected within the same local network to perform connectivity testing and troubleshooting operations.

4 Initial Network Configuration

The following IP addressing scheme was configured:

- PC0 → 192.168.10.10
- PC1 → 192.168.10.11
- PC2 → 192.168.10.12
- Default Gateway → 192.168.10.1

5 Simulated Network Issues

Two troubleshooting scenarios were created:

5.1 Incorrect Gateway Configuration

The default gateway of PC1 was intentionally changed to an incorrect address to simulate gateway misconfiguration.

5.2 Incorrect IP Address Configuration

The IP address of PC2 was changed to a different subnet, causing network communication failure between devices.

6 Troubleshooting Process

The following troubleshooting methods were used:

- Ping tests to verify connectivity
- IP configuration verification
- Identification of subnet mismatch issues
- Correction of incorrect network settings

7 Testing and Verification

After correcting the network configurations, connectivity tests were performed again. All devices successfully communicated with each other, confirming successful troubleshooting and restoration of network functionality.

8 Project Screenshots

8.1 Network Topology



8.2 IP Address Configuration

assets/ipaddress-config.png

8.3 Incorrect Gateway Configuration

assets/break1-change-gateway.png

8.4 Incorrect IP Address Configuration

assets/break2-change-ipaddress.png

8.5 Troubleshooting Process

assets/troubleshoot.png

8.6 Fixing Network Issues

`assets/fix-problem.png`

8.7 Successful Connectivity Test



`assets/test-again.png`

9 Conclusion

This project successfully demonstrates practical network troubleshooting and diagnostics skills using Cisco Packet Tracer. The lab provided hands-on experience in identifying and resolving common network configuration issues, which are essential skills for IT support and network engineering roles.

10 Tools and Technologies Used

- Cisco Packet Tracer
- TCP/IP Networking
- Ping
- IP Configuration Tools

- Network Troubleshooting Techniques